Does In-Domain Masked Autoencoding Overtake ImageNet Initialisation Under Label Scarcity? Two Pre-Registered Studies Across Four Visual Domains

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***Abstract*—For label-poor domains it is often suggested to skip ImageNet and pretrain on the domain's own unlabelled images. We tested this for masked autoencoders (MAE) and a small Vision Transformer (ViT): as labels grow, when does a ViT pretrained with in-domain MAE beat the same ViT started from ImageNet-1k weights, and does distance from ImageNet tell us in advance? Two pre-registered studies covered four domains (plant disease, satellite, objects, animals), leakage-safe splits, label budgets from 5 to 100 % and three seeds on one laptop GPU. In Study 1 (240 runs) the ImageNet ViT won all 20 dataset–budget cells. The gap closed steadily with more labels but never reversed (on PlantVillage from -0.239 to -0.020 macro-F1), and none of four domain distances predicted it (exact $p \geq 0.33$). Study 2 (405 runs) removed three possible handicaps. A four times larger unlabelled pool did not improve MAE (all differences within ± 0.02), a wider learning-rate grid left ImageNet ahead in all 20 cells, and ResNet-34 behaved like ResNet-18. Continuing MAE from the ImageNet weights was never significantly better than ImageNet alone and was significantly worse in 10 of 20 cells. At this scale, ImageNet initialisation is the safer default.**

***Index Terms*—self-supervised learning, masked autoencoders, vision transformers, transfer learning, label efficiency, domain distance, calibration, plant disease**

I. Introduction

Labels are the expensive part of most vision projects outside everyday photography. Anyone can photograph a tomato leaf, but deciding which of a dozen diseases it shows takes a plant pathologist. Satellite land-cover maps and medical scans have the same problem. So nobody trains from scratch; the usual first step is to load weights learned on ImageNet [1], a collection of 1.28 million labelled photographs of everyday objects.

Self-supervised learning offers a second option. A masked autoencoder (MAE) [2] hides most of an image's patches and trains the network to paint them back in. It needs no labels, so it can be run on exactly the images one cares about. The choice is therefore between a lot of knowledge from the wrong visual world and a little knowledge from the right one. Several papers report that MAE pretraining on small target datasets helps ViTs [3]–[6], and in-domain contrastive pretraining has been found to compete with ImageNet in some settings [7]. What we could not find is an answer to the practical question *at what label budget* the in-domain route starts to pay off, and whether one can tell in advance from how different the domain looks from ImageNet. Ericsson et al. [8] list exactly this link between domain distance and transfer as open.

We set out to answer it with a controlled, pre-registered design. Our contributions are:

1) A comparison of four starting points on four domains in which the three ViTs are architecturally identical and differ only in their initial weights, trained on the same nested label subsets with the same protocol.
2) A crossover analysis over five label budgets. In-domain MAE never overtakes ImageNet, but the gap shrinks in every domain as labels are added.
3) Four distances between each domain and ImageNet, measured before training. They disagree with one another and none of them predicts the gap.
4) A follow-up study that removes three possible excuses for MAE (too little unlabelled data, a narrow learning-rate grid, a weak convolutional baseline) and tests a hybrid that continues MAE from ImageNet weights. None of these changes the conclusion.
5) Evidence that rankings by accuracy replicate across domains, whereas rankings by calibration after temperature scaling do not.

Both studies were written down and committed to a public repository before their first training run, and together they took about 36 h on a laptop GPU.

II. Related Work

Transfer from ImageNet. Models that do better on ImageNet usually transfer better [9], but Raghu et al. [10] found that for medical images much of what ImageNet pretraining provides is not needed. ViTs [11] lack the built-in locality of convolutional networks [12] and lean heavily on pretraining; DeiT [13] made it possible to train them on ImageNet-1k alone.

Self-supervised and in-domain pretraining. Ericsson et al. [8] showed that how well self-supervised models transfer

depends strongly on the target domain. Cole et al. [7] varied the size and the domain of the pretraining set for contrastive ResNet-50 models. MAE [2] hides 75 % of the patches and regresses their pixels. El-Nouby et al. [3] and Zhou et al. [4] reported that masked pretraining on the target data alone can be competitive, and Mao et al. [5] and Das et al. [6] used MAE to make ViTs less data hungry. Hammoud et al. [14] looked at the role of diversity in the pretraining data. Continuing to pretrain an existing model on in-domain data is a well-established idea in language modelling [15], and Reed et al. [16] found that self-supervised pretraining started from ImageNet weights converges faster and transfers better than starting from scratch. On the plant side, deep networks are highly accurate on PlantVillage [17], [18], and self-supervised transformers have been trained on plant images [19]. We did not find a study that locates the label budget at which in-domain MAE overtakes ImageNet across several domains.

Dataset distance and calibration. FID [20] compares two feature distributions, and OTDD [21] also accounts for the labels. Guo et al. [22] showed that modern networks are overconfident and proposed temperature scaling; Minderer et al. [23] revisited the question for newer architectures, and Sambyal et al. [24] studied calibration with limited medical data. Dehghani et al. [25] caution that conclusions drawn on one benchmark often fail to carry over, which is why we use several domains and rank statistics [26].

III. Method

A. Arms

Every arm is fine-tuned on the same labelled subsets (Fig. 1). In Study 1 there are four. **A** is a ResNet-18 [12] trained from random weights (11.2 M parameters). **C** is a ViT-Tiny with the DeiT-Tiny layout [13] (12 blocks, width 192, 3 heads, 5.5 M parameters) loaded with supervised ImageNet-1k weights. **D** is the same ViT from random weights, and **E** is the same ViT whose weights come from an MAE trained on the domain's own training images, without their labels. Since C, D and E have one architecture, any difference between them comes from the starting weights. Our main contrast is E−C; E−D shows what MAE adds to a ViT, and E−A compares it with the inductive bias of a CNN.

Study 2 adds three arms. **E2** is E with a four times larger unlabelled pool. **F** starts from C's ImageNet weights, continues MAE pretraining on the same pool, and is then fine-tuned like the others. **A2** replaces ResNet-18 by ResNet-34 (21.3 M parameters). We kept ViT-Tiny, small ResNets and 160 px inputs so that each study fits within 48 h on one RTX 4050 Laptop GPU with 6 GB of memory.

B. Data, splits and label budgets

Each domain is represented by one dataset (Table I): PlantVillage [17] (plant disease, colour images), EuroSAT RGB [27] (land cover from satellites), CIFAR-10 [28] (objects) and Oxford-IIIT Pets [29] (cat and dog breeds). We draw 7,000 images per dataset, stratified by class, so all domains start with the same amount of data. Images are resized so that the shorter side is 160 px (bicubic) and centre-cropped; CIFAR-10 and EuroSAT are upsampled in the process.

Datasets such as PlantVillage contain many near-copies, for example several shots of the same leaf. If such copies land on both sides of a split, test scores are inflated. We therefore hash every image (256-bit difference hash), link images whose hashes differ in at most 10 bits, and keep each linked group inside one split. The 70/15/15 split is made with `StratifiedGroupKFold` (seed 42). The label budgets are 5, 10, 20, 50 and 100 % of each class's training images, with at least one image per class, which corresponds to roughly 250 to 5,000 labelled images. Smaller subsets are contained in larger ones, and all arms see the same subset for a given seed. The seed (0, 1 or 2) also fixes the order of the data, the initialisation of any new layers and the MAE run used by E.

C. Distance between a domain and ImageNet

We pass every image through the frozen ImageNet DeiT-Tiny of arm C and keep its 192-d class token. ImageNetV2 (matched frequency) [30], with 10,000 images over all 1,000 classes, stands in for ImageNet. On the training split we compute four numbers. (i) FID [20] between the two sets of features. (ii) OTDD [21], an optimal-transport distance in which the cost of matching two images also depends on how similar their classes are; we use a diagonal Gaussian per class, 64 PCA dimensions and 1,500 samples per side, solved with POT [31]. (iii) ImageNet confidence: the average top-class probability that the 1,000-way ImageNet classifier assigns to the domain's images (0.458 on ImageNetV2 itself). (iv) The validation accuracy of a logistic-regression probe trained on the frozen features. Large FID or OTDD, and small confidence or probe accuracy, mean the domain is far from ImageNet.

D. MAE pretraining

The MAE encoder is exactly the ViT-Tiny used by the other ViT arms, so its weights drop straight into E. Of the 100 patches of an image, 75 % are hidden. A light decoder (2 blocks, width 128, 4 heads) predicts the per-patch normalised pixels of the hidden patches. We use batch 256, AdamW with $\beta = (0.9, 0.95)$, a base learning rate of 1.5×10^{-4}, weight decay 0.05, a cosine schedule, random resized crops (scale 0.2–1) and horizontal flips. Encoder position embeddings start from fixed 2-D sine–cosine values. For E the pool is the ∼5,000 unlabelled training images, seen for 800 epochs (15,200 steps, 40 warm-up epochs). A separate MAE is trained for each seed, which takes about 21.5 min.

In Study 2 the pool grows to 20,000 images: the training images plus 15,000 extra images of the same dataset taken from *outside* the 7,000-image sample, so no validation or test image can enter it. As a further safeguard, we drop any extra image whose hash lies within 10 bits of a validation or test image or of a flipped copy of one (227 images on EuroSAT, none elsewhere). E2 is trained for the same 15,200 steps as E with 760 warm-up steps, so only the data changes, not the compute. F uses the same schedule but starts from the ImageNet weights, keeps their learned position embeddings

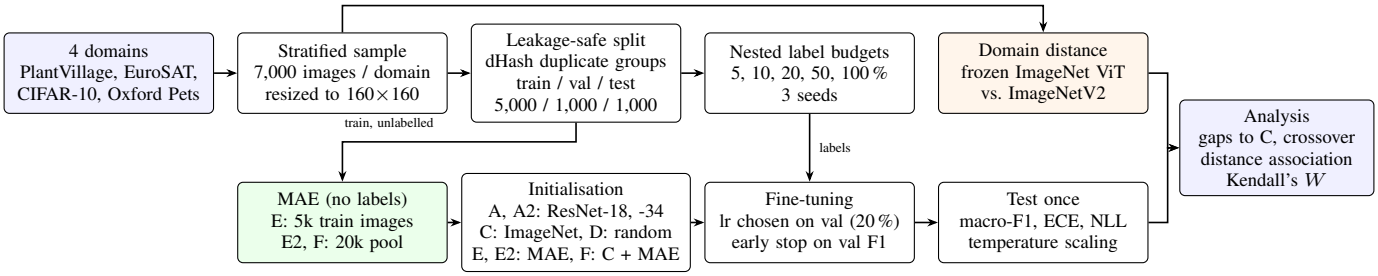


Fig. 1. Pipeline, run separately for each domain. Study 1 uses arms A, C, D and E; Study 2 adds A2, E2 and F. Only MAE pretraining sees images before fine-tuning, and never validation or test images. Each run is evaluated on the test set once.

TABLE I
DATASETS, LEAKAGE-SAFE SPLITS AND DISTANCES TO IMAGENET, MEASURED BEFORE TRAINING. THE ARROW SHOWS WHICH DIRECTION MEANS "FARTHER FROM IMAGENET".

Dataset	Domain	Classes	Native px	Train/Val/Test	Dup. groups	FID ↑	OTDD ↑	ImageNet conf. ↓	Linear probe ↓
PlantVillage	plant	38	256	4999/1000/1001	0	433	30.6	0.30	0.90
EuroSAT	satellite	10	64	5000/1000/1000	1	318	26.4	0.14	0.86
CIFAR-10	objects	10	32	5000/1000/1000	0	267	28.3	0.29	0.83
Oxford Pets	animals	37	var.	5000/1000/1000	34	273	29.3	0.51	0.86

TABLE II
FINE-TUNING PROTOCOL, SHARED BY ALL ARMS UNLESS STATED OTHERWISE.

Optimiser	AdamW [33], weight decay 0.05
Batch / precision	128 / bf16, gradient clipping 1.0
Schedule	5 % linear warm-up, cosine decay
Steps	$\mathrm{clip}(100\lceil n/128\rceil, 600, 4000)$
Early stopping	25 val. checks, patience 8, restore best
Augmentation	random resized crop 0.35–1, flip, brightness/contrast $\pm 20\,\%$
ViT details	drop-path 0.1, average-pooled tokens
Layer-wise lr decay	0.75 (pretrained ViTs: C, E, E2, F)
LR grid, scratch	S1: $\{0.5, 1, 2\} \times 10^{-3}$
	S2: $\{0.25, 0.5, 1, 2, 4\} \times 10^{-3}$
LR grid, pretrained	S1: $\{0.25, 0.5, 1\} \times 10^{-3}$
	S2: $\{0.125, 0.25, 0.5, 1, 2\} \times 10^{-3}$

and uses a smaller learning rate of 5×10^{-5} so as not to wipe out what ImageNet taught. That value was fixed in advance and not tuned. Oxford Pets has no images beyond our sample, so there is no E2 for Pets, and its F uses the 5,000 training images.

E. Fine-tuning

All arms follow Table II. The number of steps grows with the number n of labelled images. We check validation macro-F1 25 times per run, stop after 8 checks without improvement, restore the best checkpoint and only then evaluate on the test set. Pretrained ViTs use layer-wise learning-rate decay of 0.75 [32], so that the early blocks move less than the late ones. The learning rate for each dataset and arm is picked on validation macro-F1 at the 20 % budget with seed 0 and then reused for every budget and seed. Study 2 widens the grid from three to five values (Table II) and retunes every arm. Where the choice for an old arm stays the same, the Study 1 runs are reused; where it changes, those runs are repeated.

F. Metrics and statistics

Test macro-F1 is the primary metric. For calibration we report the expected calibration error (ECE, 15 equal-width bins [22], [34]) before and after temperature scaling, with the temperature fitted on validation data. Gaps between two arms are paired by seed, because both see the same labelled subset, and are given with a 95 % t-interval over the three seeds. The crossover budget is where the mean gap changes sign, found by interpolating on a log-budget axis. If the sign never changes we report the crossover as censored ($>100\,\%$ or $<5\,\%$), and we accept a crossover only if the intervals exclude zero on both sides of it. To summarise a dataset with one number we average the gap over the log-budget axis (trapezoidal rule). We relate this mean gap to each distance with Spearman's ρ, an exact permutation p-value and a leave-one-dataset-out linear prediction, compared against simply predicting the mean of the other datasets. Four datasets allow no two-sided p smaller than $2/4! = 0.083$, so we declared this part exploratory from the start. Agreement of the arm rankings across datasets is measured per budget with Kendall's W [35] and the Friedman test.

G. Pre-registration

Study 1's hypotheses (H1–H5), protocol and analysis plan were committed to the repository before the first training run. A single amendment, also made before training, raised MAE pretraining from 400 to 800 epochs once a speed test showed that the budget allowed it. Study 2's hypotheses (H6–H11) were committed after Study 1 had finished but before any Study 2 code was run. Table VIII lists all eleven with their outcomes.

IV. STUDY 1: RESULTS

All 240 runs finished: 9.3 h of fine-tuning, 1.0 h of learning-rate selection and about 4.3 h of MAE pretraining, 14.6 h in

TABLE III
STUDY 1 TEST MACRO-F1 (MEAN ± S.D. OVER 3 SEEDS). BOLD: BEST ARM. LAST COLUMN: PAIRED GAP E−C; *95 % CI EXCLUDES 0.

	%	A: CNN scr.	C: ViT IN	D: ViT scr.	E: ViT MAE	E−C
PlantVillage	5	0.548 ±0.019	**0.806 ±0.012**	0.500 ±0.038	0.567 ±0.020	−0.239*
	10	0.697 ±0.033	**0.861 ±0.039**	0.607 ±0.019	0.693 ±0.019	−0.167*
	20	0.810 ±0.016	**0.940 ±0.006**	0.700 ±0.010	0.812 ±0.024	−0.127*
	50	0.930 ±0.007	**0.979 ±0.010**	0.821 ±0.008	0.925 ±0.014	−0.054*
	100	0.970 ±0.003	**0.982 ±0.006**	0.911 ±0.005	0.962 ±0.005	−0.020*
EuroSAT	5	0.764 ±0.016	**0.934 ±0.009**	0.714 ±0.032	0.751 ±0.005	−0.183*
	10	0.813 ±0.016	**0.954 ±0.010**	0.784 ±0.004	0.799 ±0.020	−0.155*
	20	0.851 ±0.014	**0.964 ±0.004**	0.846 ±0.011	0.853 ±0.010	−0.111*
	50	0.913 ±0.010	**0.977 ±0.003**	0.895 ±0.006	0.910 ±0.007	−0.066*
	100	0.949 ±0.006	**0.982 ±0.003**	0.918 ±0.007	0.933 ±0.003	−0.049*
CIFAR-10	5	0.382 ±0.012	**0.868 ±0.020**	0.314 ±0.030	0.425 ±0.017	−0.443*
	10	0.452 ±0.026	**0.887 ±0.006**	0.365 ±0.032	0.506 ±0.004	−0.380*
	20	0.537 ±0.017	**0.904 ±0.008**	0.419 ±0.007	0.581 ±0.014	−0.323*
	50	0.674 ±0.004	**0.926 ±0.008**	0.501 ±0.005	0.672 ±0.005	−0.255*
	100	0.746 ±0.006	**0.932 ±0.010**	0.565 ±0.011	0.736 ±0.018	−0.196*
Oxford Pets	5	0.117 ±0.010	**0.771 ±0.008**	0.073 ±0.006	0.151 ±0.033	−0.620*
	10	0.179 ±0.013	**0.815 ±0.008**	0.102 ±0.013	0.210 ±0.020	−0.604*
	20	0.283 ±0.014	**0.853 ±0.007**	0.122 ±0.007	0.305 ±0.009	−0.548*
	50	0.466 ±0.001	**0.880 ±0.006**	0.196 ±0.004	0.502 ±0.018	−0.378*
	100	0.659 ±0.015	**0.890 ±0.009**	0.286 ±0.014	0.638 ±0.017	−0.252*

TABLE IV
STUDY 1 GAPS PER DATASET (MACRO-F1). MEAN GAP: E−C AVERAGED OVER THE LOG-BUDGET AXIS. RANGES RUN OVER THE FIVE BUDGETS; "SIG." COUNTS BUDGETS WHOSE 95 % CI EXCLUDES 0.

	E−C		E−D		E−A	
Dataset	mean gap	crossover	range	sig.	range	sig.
PlantVillage	−0.118	> 100 %	+0.051 to +0.113	4/5	−0.008 to +0.019	0/5
EuroSAT	−0.110	> 100 %	+0.007 to +0.037	0/5	−0.015 to +0.002	0/5
CIFAR-10	−0.317	> 100 %	+0.111 to +0.171	5/5	−0.010 to +0.055	1/5
Oxford Pets	−0.489	> 100 %	+0.078 to +0.353	4/5	−0.020 to +0.037	1/5

total, never using more than 3.4 GB of GPU memory.

A. ImageNet wins every cell and there is no crossover

Arm C had the best mean macro-F1 in all 20 dataset–budget cells (Fig. 2, Table III). The paired gap E−C was negative in every cell, and its 95 % interval excluded zero every time (last column of Table III). The crossover budget is therefore censored at >100 % for all four datasets (Table IV). H1 holds only in a weak form: the gap closes as labels are added, in every domain, but it does not flip. The closing is smooth. On PlantVillage the gap goes from -0.239 at 5 % to -0.020 at 100 % (95 % CI $[-0.037, -0.003]$), and on EuroSAT from -0.183 to -0.049. On CIFAR-10 (-0.443 to -0.196) and Oxford Pets (-0.620 to -0.252) a wide gap remains even with all labels.

B. Does distance from ImageNet predict the gap?

The mean gap was far smaller for PlantVillage (-0.118) and EuroSAT (-0.110) than for CIFAR-10 (-0.317) and Oxford Pets (-0.489). The last two are full of objects and animals that ImageNet covers well; Pets is in fact the only domain where the ImageNet classifier is more confident (0.51) than on ImageNetV2 (0.46). The trouble is that the four distances do not agree on which domain is far away (Table I). FID puts PlantVillage farthest, ImageNet confidence puts EuroSAT farthest, and OTDD puts EuroSAT *closest*. Their link to the gap is just as mixed (Table V). FID ($\rho = +0.6$) and ImageNet confidence ($\rho = -0.8$) point the way H2 predicted, OTDD points the other way ($\rho = -0.4$) and the linear probe shows nothing. No p-value is below 0.33, and only ImageNet confidence predicts a held-out dataset better than the mean does (absolute error 0.124 against 0.193). With four domains, H2 is not supported.

TABLE V
DISTANCE VERSUS MEAN GAP E−C (EXPLORATORY, $n = 4$). "PRED.": SIGN OF ρ EXPECTED UNDER H2. LOO: MEAN ABSOLUTE ERROR OF A LEAVE-ONE-DATASET-OUT LINEAR FIT, AGAINST PREDICTING THE MEAN OF THE OTHER DATASETS.

Distance	Pred.	ρ	exact p	LOO error	baseline
FID	+	+0.600	0.417	0.288	0.193
OTDD	+	−0.400	0.750	0.350	0.193
ImageNet confidence	−	−0.800	0.333	0.124	0.193
Linear-probe accuracy	−	+0.000	1.000	0.205	0.193

C. MAE against training from scratch

In-domain MAE beat the scratch ViT in all 20 cells, significantly in 13 (Table IV), so H3 is mostly supported. The benefit was largest on Oxford Pets, where it reached $+0.353$ at 100 % (0.638 against 0.286), and it was also large on CIFAR-10 ($+0.111$ to $+0.171$). On EuroSAT it was small ($+0.007$ to $+0.037$) and never significant, because the scratch ViT was already good there (0.918 at 100 %). Compared with the scratch ResNet-18, MAE stayed between -0.020 and $+0.055$, with only 2 of 20 differences significant (CIFAR-10 at 5 %, Pets at 20 %). Put simply, at this scale MAE gives a ViT about what its convolutional design gives a ResNet, and not more.

D. Do the rankings replicate?

The order of the four arms by macro-F1 was nearly the same in every domain: C first, then E and A close together, then D. Kendall's W lay between 0.90 and 1.00 at every budget (Friedman $p \leq 0.013$; Table VI). H4, which expected the winner to change between domains, is rejected.

Calibration told a different story. Before temperature scaling, C also had the lowest ECE in 17 of 20 cells (mean 0.052, against 0.119 for E, 0.154 for D and 0.171 for A). Temperature scaling brought every arm down to 0.024–0.032 on average, and after it the ranking no longer agreed across datasets ($W = 0.05$–0.53, all Friedman $p \geq 0.098$). The scratch ViT was even the best-calibrated arm in 7 cells. H5 is partly supported: calibration rankings part ways with accuracy rankings once a simple correction is applied.

V. STUDY 2: WAS STUDY 1 UNFAIR TO MAE?

A negative result invites the question of whether the losing method was given a fair chance. Study 2 checks the three objections we found most convincing and adds the hybrid F. It produced 405 runs, of which the unchanged Study 1 cells were reused, and took about 21 h; the laptop GPU ran at full load but throttled itself at ∼89 °C, so each MAE run took

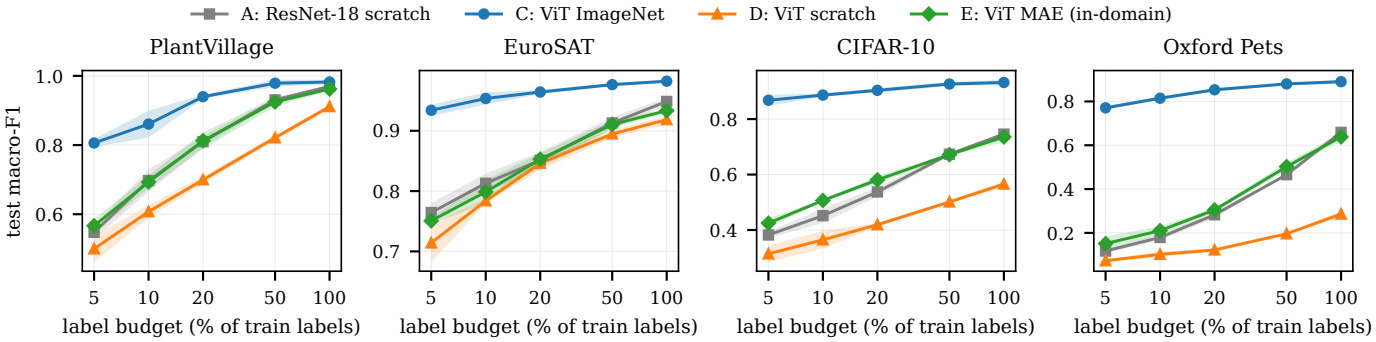


Fig. 2. Study 1: test macro-F1 against the label budget (mean of three seeds, band = ±1 s.d.). The ImageNet ViT (C) is on top everywhere. In-domain MAE (E) sits above the scratch ViT (D) and close to the scratch ResNet-18 (A).

TABLE VI
STUDY 1: AGREEMENT OF ARM RANKINGS ACROSS THE FOUR DATASETS, PER BUDGET. KENDALL'S W (1 = IDENTICAL RANKINGS) AND FRIEDMAN p.

	macro-F1		ECE		ECE after TS	
Budget %	W	p	W	p	W	p
5	0.925	0.011	0.775	0.026	0.050	0.896
10	0.900	0.013	0.900	0.013	0.075	0.825
20	1.000	0.007	0.375	0.212	0.525	0.098
50	0.925	0.011	0.700	0.038	0.450	0.145
100	1.000	0.007	0.550	0.086	0.225	0.440

22.8 instead of 21.5 min. Table VII and Fig. 3 summarise the outcome.

A. A wider learning-rate grid

With five candidate rates instead of three, the chosen rate changed for 9 of the 16 Study 1 dataset–arm pairs, and those 135 runs were repeated. The picture did not move. C still won all 20 cells among the original four arms (H11 supported), and E−C stayed negative in all 20, significantly so in 19. The closest call is again PlantVillage at full labels, now −0.012 instead of −0.020.

B. Four times more unlabelled images

If MAE simply had too little data, a bigger pool should have helped. It did not. Across the three datasets that had extra images, E2 and E differed by at most 0.02 macro-F1 in any cell, E2 was ahead in 8 of 15 cells, and none of the differences was significant (H6 not supported). E2 also stayed below C in all 15 cells, 14 of them significantly, and its best result was still 0.014 short (PlantVillage at 100 %; H7 not supported). Even the MAE reconstruction loss barely changed: after the same number of steps it ended at 0.407 rather than 0.409 on PlantVillage and 0.266 rather than 0.268 on CIFAR-10.

C. ImageNet first, then in-domain MAE

The hybrid F was meant to take the best of both. It did not. F beat C in only 3 of 20 cells, by at most 0.006 and never significantly, and it was significantly worse in 10 (H8 not supported). The damage is concentrated at small budgets. At 5 % labels F fell behind C by 0.351 on CIFAR-10 and by 0.542 on Pets, which is most of the advantage ImageNet had over in-domain MAE. With all labels F recovered to within 0.04 of C everywhere. As expected, F was well ahead of E2 (by 0.021 to 0.210, significant in 14 of 15 cells; H9 supported), but that edge comes from the ImageNet start, not from the extra MAE. F also behaved more like a model trained from scratch: it stopped early in 17 % of its runs, close to E (20 %) and far from C (67 %). Its correlation with the domain distances repeated the pattern of Study 1, with no p below 0.33.

D. A larger ResNet

ResNet-34 and ResNet-18 were within 0.03 of each other in 18 of 20 cells. The only significant difference was on CIFAR-10 with all labels (+0.042). Swapping the CNN reversed the comparison with the best MAE arm in a single cell, so H10 is partly supported, and the claim that MAE roughly matches a scratch CNN at this scale stands. As for calibration, the mean ECE before temperature scaling was 0.052 for C, 0.116 for E2, 0.146 for F, 0.151 for E and 0.176 for ResNet-34; after scaling all arms fell between 0.024 and 0.033, as in Study 1.

VI. DISCUSSION

Why ImageNet keeps winning. ImageNet-1k holds about 250 times more images than Study 1's in-domain pool and 64 times more than Study 2's, and its labels push the network to pick up shapes, parts and textures. These generic features turn out to be useful even for leaves and satellite tiles. The ImageNet ViT also learned fastest during fine-tuning and triggered early stopping in two thirds of its runs. MAE, in contrast, learns by repainting pixels. Its reconstructions (Fig. 4) get the overall layout right, such as the outline and colour of a leaf, but blur the small details, including the lesions that separate many PlantVillage classes. That would explain why the gap shrinks as labels are added: with enough labels, fine-tuning can learn the fine detail that the pretraining objective ignored.

Why more unlabelled data did not help. With 20,000 instead of 5,000 images and the same number of steps, MAE reached the same reconstruction loss and the same downstream accuracy. At this model size and schedule, the bottleneck

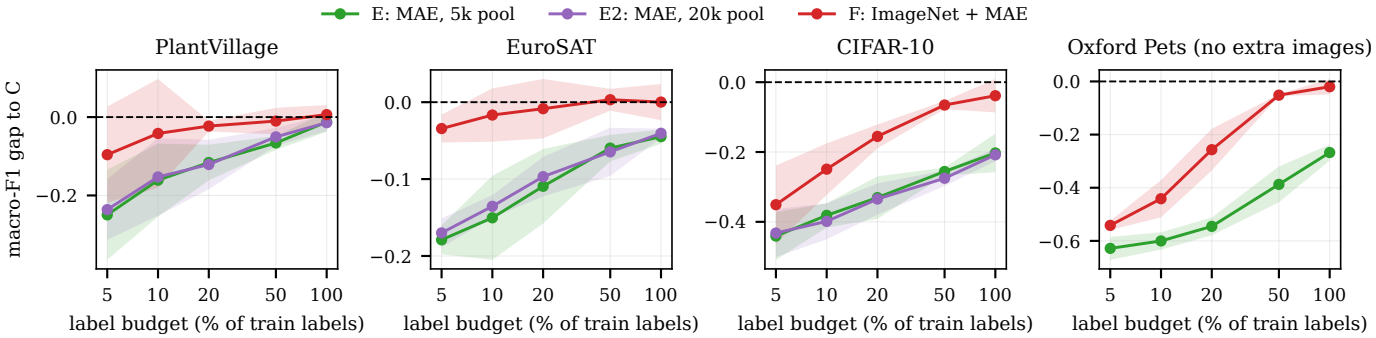


Fig. 3. Study 2: paired macro-F1 gap to the ImageNet ViT (C) for in-domain MAE on 5,000 images (E), on 20,000 images (E2), and for ImageNet weights with continued MAE (F), with 95 % t-intervals over three seeds. E2 lies on top of E. F starts far below C and catches up only when labels are plentiful.

TABLE VII
STUDY 2 PER DATASET, AFTER RETUNING ON THE WIDER LEARNING-RATE GRID. MEAN GAPS ARE AVERAGED OVER THE LOG-BUDGET AXIS (MACRO-F1). "SIG. +/−": BUDGETS WHOSE 95 % CI LIES ENTIRELY ABOVE / BELOW 0. "F>C / SIG. −": BUDGETS WHERE F'S MEAN IS HIGHER THAN C'S / BUDGETS WHERE F IS SIGNIFICANTLY LOWER.

	E−C	E2−E (more unlabelled data)		E2−C	F−C	F>C / sig. −	A2−A (ResNet-34 − 18)
Dataset	mean gap	mean gap	sig. +/−	mean gap	mean gap	(of 5 budgets)	range over budgets
PlantVillage	−0.116	+0.006	0/0	−0.110	−0.029	1/1	−0.015 to +0.009
EuroSAT	−0.106	+0.007	0/0	−0.099	−0.009	2/1	−0.008 to +0.003
CIFAR-10	−0.320	−0.010	0/0	−0.330	−0.162	0/4	−0.002 to +0.042
Oxford Pets	−0.493	–	–	–	−0.250	0/4	−0.011 to +0.034

TABLE VIII
ALL PRE-REGISTERED HYPOTHESES AND THEIR OUTCOMES. H1–H5 WERE REGISTERED BEFORE STUDY 1, H6–H11 BEFORE STUDY 2.

	Hypothesis	Outcome
H1	C ahead at small budgets; E closes or reverses the gap at larger budgets	**Partly**: the gap closes in all 4 domains, but no crossover up to 100 %
H2	Gap E−C smaller for domains farther from ImageNet	**Not supported**: right direction for 2 of 4 distances, none significant
H3	E > D at every budget in every domain	**Mostly**: 20/20 positive, 13/20 significant, none of them on EuroSAT
H4	The best arm differs across domains and budgets ($W < 1$)	**Rejected**: C best in 20/20 cells, $W = 0.90$–1.00
H5	Ranking by ECE differs from ranking by macro-F1	**Partly**: similar before temperature scaling, different after
H6	A 20,000-image pool improves MAE (E2 > E) on PlantVillage, CIFAR-10 and EuroSAT	**Not supported**: differences within ±0.02, 8/15 positive, none significant
H7	E2 narrows the gap to C at every budget and overtakes C somewhere	**Not supported**: E2 below C in 15/15 cells (14 significant)
H8	ImageNet + MAE (F) beats C in most cells, more so for distant domains	**Not supported**: F > C in 3/20 (none significant), significantly worse in 10/20
H9	F ≥ E2 on the three datasets with a pool	**Supported**: F ahead by 0.021–0.210, 14/15 significant
H10	ResNet-34 within 0.03 of ResNet-18 in most cells; MAE-vs-CNN verdict unchanged	**Partly**: within 0.03 in 18/20; verdict flips in 1 cell
H11	The wider lr grid does not change the winner among A, C, D, E	**Supported**: C still best in 20/20 cells

seems to be what pixel reconstruction teaches rather than how many images it sees. Data volumes in the MAE literature are often orders of magnitude larger [2], so our result says "four times more does not help", not "more can never help".

Why continued MAE hurt. Starting MAE from ImageNet weights moved the network away from the class-level features that made C strong, towards features that are good for repainting pixels. With few labels there is not enough signal to pull it back, which is why the damage is largest at 5–20 % and almost gone at 100 %. Two caveats apply. Reed et al. [16] saw benefits from such hierarchical pretraining, but they started from self-supervised weights and used contrastive objectives, not supervised weights and pixel reconstruction. And our MAE learning rate for F was a fixed guess: F's reconstruction loss stayed above that of the MAE trained from scratch (0.456 against 0.407 on PlantVillage), so a gentler or shorter schedule might behave differently. What we can say is that this obvious recipe is not a free improvement.

Where in-domain MAE comes closest. The gap was smallest for the two domains that look least like ImageNet's object photographs (PlantVillage and EuroSAT) and largest where the content overlaps with ImageNet classes. That fits the

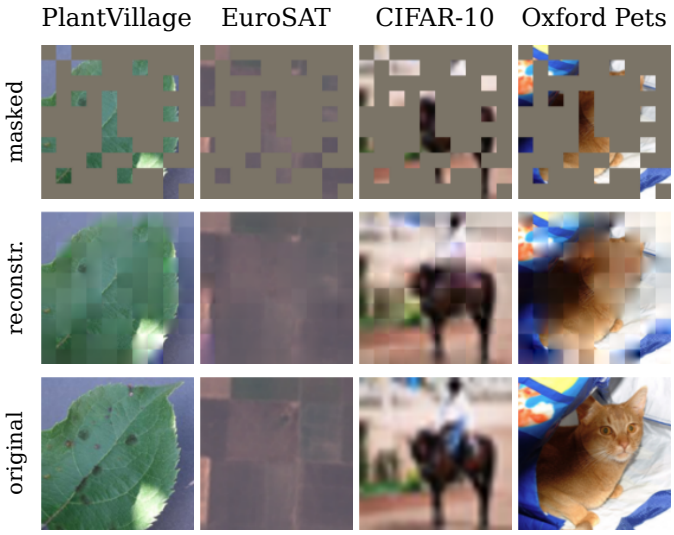


Fig. 4. MAE reconstructions from Study 1 (seed 0, one training image per domain). Top: input with 75 % of the patches hidden; middle: reconstruction; bottom: original. The layout comes back, fine detail such as leaf lesions does not.

intuition behind H2, yet none of the four distances captured it, so a single number seems too blunt to forecast a crossover. EuroSAT adds a warning: it had by far the lowest MAE reconstruction loss (0.19, against 0.27–0.54 elsewhere) and the smallest gain over a scratch ViT. A low reconstruction loss is not, by itself, a sign of a useful representation.

What a practitioner can take from this. With a few thousand to twenty thousand unlabelled images and a small ViT, start from ImageNet, at every label budget. In domains unlike ImageNet the remaining advantage at full labels can be small (about one macro-F1 point on PlantVillage), and in-domain MAE is still a clear step up from a ViT trained from scratch. Calibration should be checked for each domain, because the accuracy ranking does not carry over to it.

VII. Limitations

The studies were sized for one laptop GPU, and that sets their limits. (i) We used ViT-Tiny at 160 px and at most 20,000 unlabelled images per domain. Bigger models, higher resolution and much bigger pools favour MAE in the literature [2], so a crossover may exist beyond our range. (ii) With four domains no rank correlation can reach $p < 0.083$, so H2 was only explored. (iii) Three seeds leave wide intervals for small effects. (iv) Even on the wider grid, 16 of the 27 chosen learning rates sit at one of its two ends, so some arms might still gain from further tuning; since the conclusions hardly moved when the grid was widened, we do not expect this to reverse them. (v) MAE used one recipe without a search over the masking ratio or the schedule, and F's MAE learning rate was not tuned. (vi) The DeiT weights were trained at 224 px and their position embeddings were interpolated to 160 px; CIFAR-10 and EuroSAT were upsampled from 32 and 64 px. (vii) Hashing cannot find every near-duplicate, for example the same leaf photographed from another angle. (viii) ImageNetV2 replaces ImageNet in the distance measures, Oxford Pets had no extra images for E2, and C benefits from far more pretraining compute than any MAE arm; we did not equalise compute between arms.

VIII. Conclusion

We asked when in-domain MAE pretraining overtakes ImageNet initialisation for a small ViT as more labels become available. Over four domains and two pre-registered studies, the answer is that it does not, anywhere between 5 and 100 % of the labels. ImageNet initialisation won all 20 settings. Its lead shrank as labels were added and was smallest for plant-disease and satellite images, but none of four standard distances predicted where it would be small. Giving MAE four times more unlabelled data, widening the learning-rate grid and using a deeper ResNet as the baseline did not change this, and continuing MAE from ImageNet weights made things worse when labels were scarce. In-domain MAE did reliably improve a ViT trained from scratch and matched a CNN trained from scratch. The natural next steps are much larger unlabelled pools, larger ViTs and more domains, which would also make it possible to test the link between domain distance and the crossover budget with real statistical power.

Reproducibility

Code, configurations, both pre-registrations, split files, every run's metrics and the experiment logs are available at https://github.com/namratapathak0410-pixel/Label-Efficient-Representation-Learning-SSL. Models were built with PyTorch and timm [36]. Every figure and table in this paper is produced from the result files by `paper/make_figures.py`.

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